# Questions: Introduction to logic

Jessica Taberner


#### Summary

A selection of questions for the study guide introducing propositional logic.

*Before attempting these questions, it is highly recommended that you read Guide: Introduction to logic.*


## Q1 Propositions

Boole's Burgers has recently opened. To help Mary and George with their advertising strategy, determine which of the following sentences are propositions.

**1.1.** "Boole's Burgers is a burger restaurant."

**1.2.** "Our burgers taste amazing!"

**1.3.** "All burgers in the restaurant cost more than £2."

**1.4.** "Please give me a cheeseburger."

**1.5.** "What are your burger toppings?"

**1.6.** "Some customers prefer vegan burgers."

**1.7.** "Are the fries sold out?"

**1.8.** "The fries are sold out."

## Q2 Connectives

What is the logical structure for the following statements?

**2.1.** "If you get two burgers, you'll get a discount."

**2.2.** "I'll get one cheeseburger and one vegan burger."

**2.3.** "I won't get cheese on my burger."

**2.4.** "I'll get a hamburger, or I'll get a milkshake."

**2.5.** "I'll only get a cheeseburger if you get one too."

**2.6.** "If you go to Boole's Burgers, I'll go too, and I'll get a vegan burger."

## Q3 Truth tables

Give the truth tables for the following logical statements:

**3.1.** $P \to Q$

**3.2.** $\neg P$

**3.3.** $(P \lor Q) \land \neg Q$

**3.4.** $\neg(P \leftrightarrow Q)$

**3.5.** $\neg(P \land Q)$

**3.6.** $(P \to Q) \to R$

**3.7.** $\neg(P \lor (Q \lor R))$

**3.8.** $(P \leftrightarrow Q) \lor (\neg R)$

## Q4 Equivalence

Using truth tables or otherwise, are the following pairs of statements logically equivalent?

**4.1.** $\neg(P \land Q)$ and $(\neg P) \lor (\neg Q)$

**4.2.** $P \to Q$ and $Q \to P$

**4.3.** $P \to Q$ and $(\neg P) \lor Q$

**4.4.** $P \to Q$ and $\neg(P \lor Q)$

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After attempting the questions above, please click this link to find the answers.

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## Version history and licensing

v1.0: initial version created 03/26 by Jessica Taberner as part of a University of St Andrews VIP project.